1. Introduction and scope

1.1 The purpose of this document is to provide guidance for the production, control, storage and distribution of excipients used in pharmaceutical products, focusing on good manufacturing practices (GMP) under an appropriate system for managing quality. It is also intended to help ensure that such excipients meet the requirements for quality and purity that they purport or are represented to possess. 1.2 The document does not cover aspects of protection of the environment, or safety aspects for the personnel engaged in the manufacture and control of excipients. 1.3 Excipients are often used in large quantities in industrial chemistry, as well as in the food and cosmetic industry. Specifications for excipients used in these applications may vary and may not always be appropriate for use in pharmaceutical products. It is the responsibility of the finished product manufacturer and of the applicant to ensure that the finished product is manufactured using excipients of a suitable grade conforming to its intended use. 1.4 Excipients should be of appropriate quality, as they could affect the safety, quality and efficacy of finished pharmaceutical products. 1.5 The manufacturer of the finished pharmaceutical product is highly dependent on the excipient manufacturer to provide materials that are homogeneous in chemical and physical characteristics, and of the desired quality. 1.6 In general, excipients are used as purchased, with no further refining or purification. Consequently, impurities present in the excipient will be carried over to the finished pharmaceutical product. 1.7 To achieve the objective of ensuring that excipients used in pharmaceutical products are of appropriate quality, an appropriate level of GMP should be established, implemented and maintained during their production, packaging, repackaging, labelling, quality control, release, storage, distribution and other related activities. Additional measures should be taken when manufacturing excipients for which scientific literature, information in the public domain or historical data indicate the presence of higher risk due to potential formation of toxic impurities during manufacturing or contamination during storage and distribution. 1.8 Manufacturers of excipients for pharmaceutical use should have a specific analytical testing procedure to ensure suitability for its intended use. Pharmacopoeial and regulatory requirements should be considered by the manufacturers as a reference for these analytical tests. Information in the public domain should also be considered. Risk management principles should be implemented in order to identify and mitigate risks. 1.9 A thorough knowledge and understanding of the processes and associated risks are required. This includes all unit operations and processing steps, including key steps in the process, critical parameters (such as time, temperature or pressure), the use of recovered solvent or mother liquor, environmental conditions, equipment used, protection from contamination and monitoring points

3. Quality management

3.1 Manufacturers involved in the production, control, storage and distribution of excipients for pharmaceutical use should establish, document, implement and maintain a comprehensively designed and clearly defined quality management system. 3.2 Senior management should assume responsibility for the quality management system, as well as the quality of the excipients for pharmaceutical use manufactured, controlled, released, stored and distributed 3.3 The quality management system should encompass the quality policy, organizational structure, procedures, processes and resources. All parts of the quality management system should be adequately resourced and maintained. 3.4 The quality management system should cover all activities necessary to ensure that excipients for pharmaceutical use will meet their intended specifications, including quality and purity. 3.5 The quality management system should incorporate the principles of good practices, which should be applied to the life cycle stages of excipients for pharmaceutical use. This includes steps such as the receipt of raw materials, production, packaging, testing, release, storage and distribution. 3.6 All quality-related activities and procedures should be defined and documented manually or electronically. 3.7 All quality-related activities should be recorded at the time they are performed. 3.8 The quality management system should ensure that: ■ sufficient resources are available (for example, equipment, personnel, materials); ■ excipients for pharmaceutical use are produced, controlled, stored and distributed in accordance with the recommendations in this document and other associated guidelines, such as good quality control laboratory practices and good storage and distribution practices, where appropriate; ■ managerial roles, responsibilities and authorities are clearly specified in job descriptions; ■ operations and other activities are clearly described in a written form, such as standard operating procedures and work instructions; ■ appropriate arrangements are made for the manufacture, supply and use of the correct containers and labels; ■ all necessary controls are in place; ■ calibrations, verification or validations are carried out where necessary; ■ the excipient for pharmaceutical use is correctly processed and checked according to the defined procedures and specifications; ■ deviations, suspected product defects, out-of-specification test results and any other nonconformances or incidents are reported, investigated and recorded. An appropriate level of root cause analysis is applied during such investigations and the most likely root causes are identified; ■ proposed changes are evaluated and approved prior to implementation. After implementation of a change that could impact the quality or the product, an evaluation should be undertaken to confirm that the quality objectives were achieved and that there was no unintended adverse impact on product quality; ■ appropriate corrective actions and preventive actions, as well as checks on the effectiveness of those actions (where appropriate), are identified and taken; ■ where required, processes are in place to ensure the management of any outsourced activities that may impact product quality, purity and integrity; ■ a batch of an excipient for pharmaceutical use is not released and supplied before it has been released by the quality unit with the assurance that the batch has been produced and controlled in accordance with product specifications, and with the recommendations in this document and any other regulations relevant to the production, control and release of these products; ■ there is a system for handling complaints, returns and recalls; ■ there is a system for self-inspection; ■ there is a system for product quality review. 3.9 The quality unit should be independent of production. The responsibilities of the unit should be clearly defined and documented. 3.10 The person or persons authorized to release excipients for pharmaceutical use should have appropriate qualifications, and be specified.

3.1 Quality risk management

3.11 There should be a system for managing risks (1). The system for quality risk management should be comprehensive and should cover a systematic process for the assessment, control, communication and review of risks in the production, testing, packaging, storage and distribution of excipients for pharmaceutical use. Controls identified should be appropriate, ensure that risks are eliminated or mitigated, and ultimately protect the patient from receiving a pharmaceutical product containing the wrong, contaminated or unsuitable excipients for pharmaceutical use. 3.12 In order to perform an adequate excipient risk assessment, it would be useful to provide some high-level guidance using an appropriate risk profile or ranking using a question-based risk ranking and filtering system. For example: ■ functionality of excipient in formulation ■ route of administration ■ potential for contamination ■ excipient complexity ■ prior knowledge or experience with excipient ■ packaging size. 3.13 Similarly, a risk score should be calculated for the supply chain (for example, complexity of supply chain, prior knowledge of supply chain, excipient manufacturer performance history, packaging suitability, quality management system standard and certification)

14. Production

14.1 Raw materials for manufacturing of excipients for pharmaceutical use should be weighed or measured in appropriate areas, under appropriate conditions, using suitable devices. 14.2 The material to be used in production should be kept in suitable containers bearing labels with required details, such as the name of the material and a traceable control number. 14.3 Equipment in production areas should be labelled, for example, with an asset or other unique identification number and, if applicable, calibration status. 14.4 Where appropriate, materials should not be kept for periods longer than the validated hold time. 14.5 The extent, stringency and type of testing (for example, in-process), as well as acceptance criteria, should be defined. All tests and results should be fully documented as part of the batch record. No. 1052, 2024 14.6 The sampling process should not increase the risk of contamination of the material. Samples should be handled with care and their integrity maintained. 14.7 Manufacturers should have written procedures and related documents for the production and control of excipients for pharmaceutical use. 14.8 Batches should be produced following written procedures or instructions. 14.9 The manufacturing process should be described in detail, and risks associated with the production and control of the excipient for pharmaceutical use should be appropriately controlled. This includes requirements specified in the recognized pharmacopoeia, transmissible spongiform encephalopathy (TSE) or bovine spongiform encephalopathy (BSE), and impurities. 14.10 Batches should be produced using suitable equipment in an appropriate environment, and should be protected from possible contamination and cross-contamination. 14.11 In-process sampling and testing should be done in accordance with written instructions. Records should be maintained. 14.12 Checks and maintenance operations should not affect the quality of the excipient for pharmaceutical use. 14.13 Changes and deviations in production should be managed through the relevant procedures. 14.14 Blending operations should be controlled to ensure homogeneity of the f inal batch. A blended batch should be assigned a unique batch number, and batches used in the blend should be traceable. 14.15 A sampling procedure should be followed to ensure that a sample collected from the blend is representative of the batch. 14.16 Each batch of product to be mixed should be produced in accordance with the batch manufacturing document, be tested separately, and meet the corresponding specifications. The mixed batch should be tested and should be in compliance with its specification. 14.17 Blending or mixing of batches should be controlled and validated. Procedures and records should be maintained. Blending of batches to salvage out-of-specification batches or adulterated material is not an acceptable practice. 14.18 Manufacturers should regularly review the capability of the process and ensure batch-to-batch consistency of the excipient for pharmaceutical use, meeting its specification. 14.19 Written procedures should be followed for the receipt, identification, quarantine, sampling, examination or testing, and release/rejection, and handling of packaging and labelling materials. Records should be kept. 14.20 Packaging materials such as containers should provide adequate protection against deterioration or contamination of the excipient for pharmaceutical use. They should be clean and dry, and should not be reactive, additive or absorptive. 14.21 Printed packaging materials such as labels should be in the prescribed format. 14.22 Access to printed packaging material storage areas should be controlled. 14.23 Stock should be reconciled at periodic intervals, including received, issued, and returned quantities. Discrepancies found should be investigated. 14.24 Batch coded labels not used for the specified batch, and obsolete and outdated labels, should be destroyed. Reconciliation should be done. Records should be maintained. 14.25 Written procedures should be followed for packaging operations. Controls should be in place to prevent any mix-ups during packaging. T hese should include line opening and line closing checks, segregation between packaging lines, and verification of materials on the packaging line prior to the start of packaging.

14.1 Rework

14.26 Reworking should only be undertaken when the outcome of a risk assessment indicates that this is acceptable and approved by the quality unit. 14.27 Batches that have been reworked should be subjected to appropriate quality control testing and stability testing, if required. A reworked batch should be released by the quality unit when it has been determined, by applying the relevant analytical testing procedures, that the specification has been met. 14.28 Records should be maintained.

14.2 Reprocessing

14.29 Reprocessing should only be undertaken if this activity and process have been evaluated internally and found to be acceptable. 14.30 Records should be maintained.